

Fig. 1: The transmitter block schematic

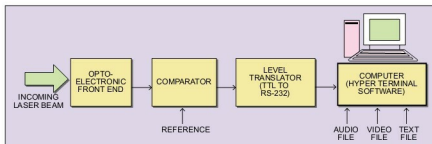


Fig. 2: The receiver block schematic



Fig. 3: Laser communication for naval applications

converted into a serial bit stream in RS/EIA232 format with the help of hyper terminal software that comes with MS Windows and is available on the PC.

In the next step, this serial bit stream (representing data) is converted into TTL compatible pulses with the help of integrated circuit (IC) MAX232. This is a dual EIA232 driver/receiver. It has two on-chip independent EIA232-to-TTL/CMOS drivers and two on-chip independent TTL/CMOS-to-EIA232 receivers. On the transmitter side, one of the receivers has been used to convert the EIA232 bit stream into equivalent 5-volt TTL/CMOS-compatible pulses.

The TTL-compatible output pulses are then fed to a non-inverting adder circuit configured around the low-noise op-amp LM324. The other input to the adder is a DC voltage, which can be adjusted from approximately 0.5

volt to 1.7 volts. The magnitude of TTL pulses is also adjustable. This is required to get the desired magnitude of the drive current through the laser diode. This circuit is part of the overall laser diode driver circuit.

The output of this circuit feeds a constant current source driver circuit configured around the op-amp LM324 and an npn Darlington transistor. The magnitude of constant current is given by the voltage appearing at terminal 3 of this op-amp and the value of resistance R_E .

The laser source used is a diode pumped solid state laser operating at 532 nm (green) with an output power of 10 mW. Green laser is used for underwater laser communication links, as it has the least attenuation in water. The proposed high-power green laser, which combines high power and high brightness with low signal noise at high repetition rates, has a great potential.

The current drive to the laser varies between two values corresponding to the low and high levels of the TTL pulse stream. As a result, the laser beam gets intensity modulated, i.e., its intensity or power gets modulated. It is a type of pulse amplitude modula-

tion. The modulated laser beam is transmitted towards the receiver. The laser communication needs line-of-sight conditions, i.e., the transmitter and receiver should be in line-of-sight.

Receiver

The modulated laser beam falls on the photo diode (type FND100). FND100 is a silicon PIN photo diode. The corresponding current signal generated by the photo diode is converted into an equivalent voltage signal, which is then amplified in a non-inverting amplifier circuit configured around op-amp AD829—a high-speed, high-bandwidth op-amp.

The amplified signal is then fed to a voltage comparator configured around op-amp comparator LM319, which is a high-speed dual comparator. One of the two comparators available on the chip has been used here. The comparator output is a TTL compatible pulse stream representing data.

This pulse stream is then fed to a TTL buffer of non-inverting type (SN 74LS244). 74LS244 is a hex non-inverting buffer. Three buffers are used in parallel for higher current drive capability. The buffer output feeds the TTL/CMOS input of one of the drivers in the MAX232 converter. The corresponding EIA232 output feeds the PC through the serial port. The hyper terminal software in the PC converts the EIA232 bit stream back to a data file, which is stored in the memory and is available on the desktop.

Laser communication—advantages and applications

Fibre-optic communication can be implemented for a longer distance underwater as compared to a laser system since distortion is much higher underwater than in free space, but underwater laser communication gives us the distinct advantage of flexibility. Fibre-optic communication can only be done where the optical fibre communication setup is available and installed. Two submarines can easily transfer data anywhere in real time